

Infrared

One of the many uses of Infrared technology is short range communication. It uses an infrared spectrum of light which is invisible to the naked eye. The device is fitted with infrared light-emitting diodes (LEDs) to emit infrared radiation which is focused by a plastic lens into a narrow beam. The beam is modulated, i.e. switched on and off, to encode the data. The receiver uses a silicon photodiode to convert the infrared radiation to an electric current. It responds only to the rapidly pulsing signal created by the transmitter, and filters out slowly changing infrared radiation from ambient light. It doesn't penetrate walls and so can't interfere with other devices in adjoining rooms. Before the Bluetooth technology, mobiles exchanged data using infrared but now the most common application of infrared technology is your everyday remote control.

Wi-Fi

Although the Wi-Fi technology has been around since 1985, it was widely used only after 1997. Wi-Fi or Wireless Fidelity is actually a play on words with Hi-Fi. It's a trademark of the Wi-Fi Alliance and the brand name for products using the IEEE 802.11 family of standards.



The yin-yang Wi-Fi logo indicates the certification of a product for interoperability

Wi-Fi has made it cheaper and easier to create local area networks in places where cables cannot be run. As the price of chipsets for Wi-Fi continues to drop, manufacturers are building wireless network adapters not only into laptops but also into cell phones and other handheld devices.

Like all wireless technologies, Wi-Fi uses electromagnetic waves to communicate. They are transmitted at frequencies of 2.4 GHz or 5 GHz. This frequency is considerably higher than the frequencies used for cell phones, walkie-talkies and televisions. The higher frequency allows the signal to carry more data and thus make it faster than Bluetooth or Infrared.

Wi-Fi uses 802.11 networking standards, which come in several variants:

- ▶ **802.11a:** Released in September 1999, it transmits at 5 GHz and can move up to 54 megabits of data per second. It also uses a more efficient